

MODULE 01

Introduction to AI and Data Science

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Introduction to AI and Data Science

What is Artificial Intelligence?

AI is the simulation of human intelligence in machines. It includes reasoning, learning, problem-solving, perception, and language understanding.

AI Types:

Narrow AI: Designed for a specific task (chess, image recognition)

General AI: Human-level reasoning across domains (future goal)

Super AI: Beyond human intelligence (theoretical)

What is Data Science?

Data Science extracts insights from structured and unstructured data using statistics, mathematics, and programming.

Data Science Pipeline:

1. Problem Definition
2. Data Collection
3. Data Cleaning
4. Exploratory Analysis
5. Model Building
6. Evaluation
7. Deployment

AI vs ML vs Deep Learning

AI: Broad concept - any machine mimicking intelligence

ML: Subset of AI - learns from data without explicit programming

DL: Subset of ML - uses neural networks with many layers

Relationship: AI contains ML contains Deep Learning

Key Applications

Healthcare: Disease diagnosis, drug discovery

Finance: Fraud detection, algorithmic trading

Transport: Self-driving cars, route optimization

Retail: Recommendation systems, demand forecasting

NLP: Chatbots, translation, sentiment analysis

Career Paths in AI/DS

Data Analyst: SQL, Excel, Tableau, Power BI

Data Scientist: Python, ML, Statistics, Modeling

ML Engineer: MLOps, model deployment, APIs

Data Engineer: Pipelines, ETL, Spark, Kafka

AI Researcher: Deep learning, novel architectures